In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_mat
```

In [3]:
```
df = pd.read_csv("weather-check.csv")
df.head()
```

Out[3]:

	RespondentID	Check_Weather_Daily	Check_Weather_Method	Weather_Smartwatch_Li
0	3887201482	Yes	The default weather app on your phone	\
1	3887159451	Yes	The default weather app on your phone	\
2	3887152228	Yes	The default weather app on your phone	\
3	3887145426	Yes	The default weather app on your phone	Somew
4	3887021873	Yes	A specific website or app (please provide the ...	\

1. Are there any null values in the dataset?

In [4]:
```
df.isnull().sum()
```

Out[4]:
```
RespondentID                    0
Check_Weather_Daily             0
Check_Weather_Method            0
Weather_Smartwatch_Likelihood   0
Age                             0
Gender                          0
Household_Income                0
US Region                       0
dtype: int64
```

2.Are there any duplicate records in the dataset?

In [5]:
```
df.duplicated().sum()
```

Out[5]:
```
0
```

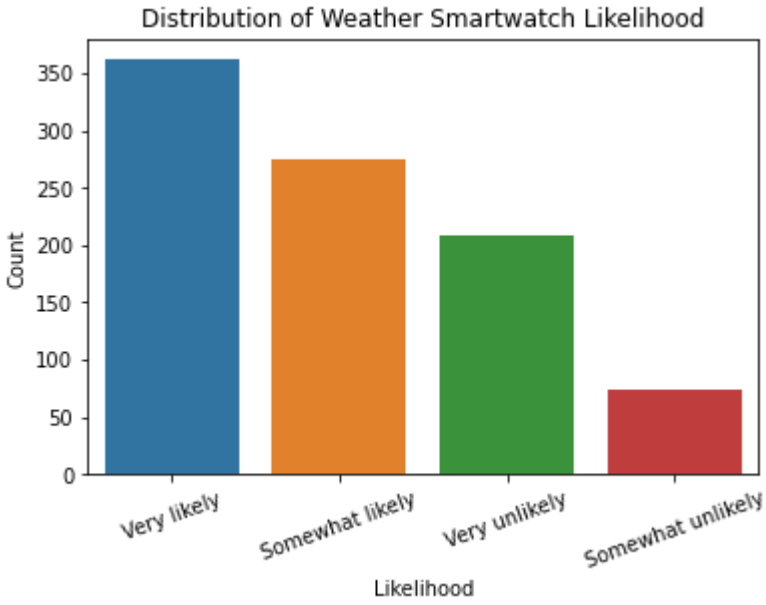
3. How many values are there in each target class?

In [6]:
```
print(df['Weather_Smartwatch_Likelihood'].value_counts())
```

```
Very likely          362
Somewhat likely      275
Very unlikely        209
Somewhat unlikely     73
Name: Weather_Smartwatch_Likelihood, dtype: int64
```

In [7]:
```
sns.countplot(data=df, x='Weather_Smartwatch_Likelihood')

plt.title("Distribution of Weather Smartwatch Likelihood")
plt.xlabel("Likelihood")
plt.ylabel("Count")
plt.xticks(rotation=20)
plt.show()
```



4. Which columns are present in the dataset?

In [8]:
```
df.columns
```

Out[8]:
```
Index(['RespondentID', 'Check_Weather_Daily', 'Check_Weather_Method',
       'Weather_Smartwatch_Likelihood', 'Age', 'Gender', 'Household_Income',
       'US Region'],
      dtype='object')
```

5. What is the summary of the dataset?

In [9]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 919 entries, 0 to 918
Data columns (total 8 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   RespondentID                    919 non-null    int64
 1   Check_Weather_Daily             919 non-null    object
 2   Check_Weather_Method            919 non-null    object
 3   Weather_Smartwatch_Likelihood   919 non-null    object
 4   Age                             919 non-null    object
 5   Gender                          919 non-null    object
 6   Household_Income                919 non-null    object
 7   US Region                       919 non-null    object
dtypes: int64(1), object(7)
memory usage: 57.6+ KB
```

6. What are the statistical characteristics of the dataset?

In [10]:
```
df.describe()
```

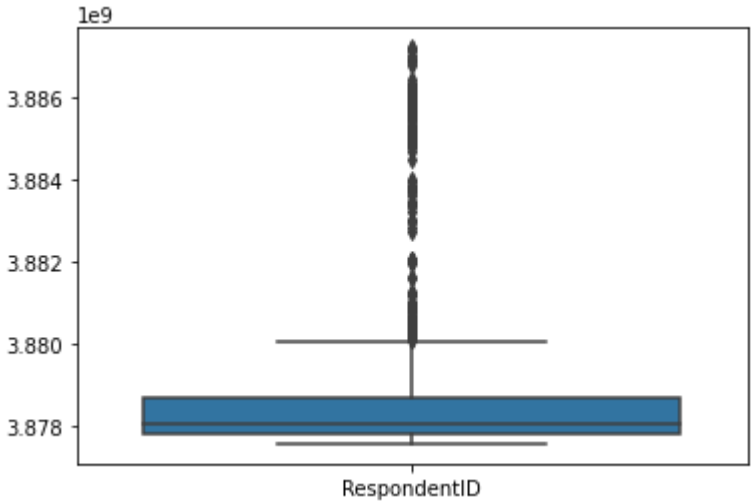
Out[10]:

	RespondentID
count	9.190000e+02
mean	3.879084e+09
std	2.445434e+06
min	3.877567e+09
25%	3.877793e+09
50%	3.878059e+09
75%	3.878700e+09
max	3.887201e+09

7. Are there any outliers in the dataset?

In [12]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(data=df)
plt.show()
```

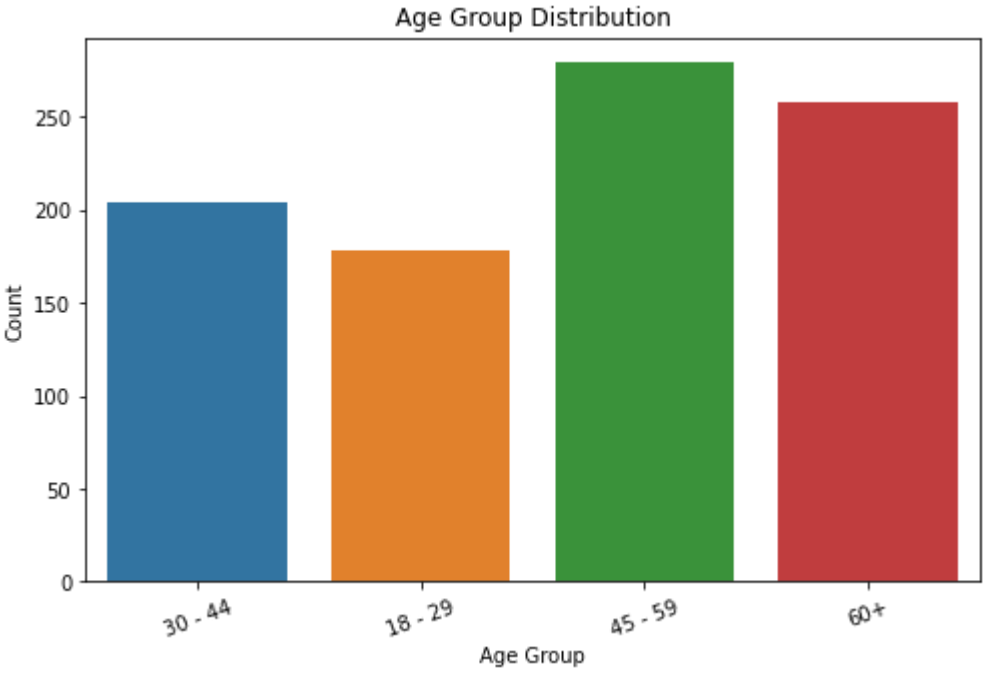


8. Are there any outliers in a particular column?

In [13]:
```
plt.figure(figsize=(8, 5))

sns.countplot(data=df, x='Age')

plt.title("Age Group Distribution")
plt.xlabel("Age Group")
plt.ylabel("Count")
plt.xticks(rotation=20)
plt.show()
```



9. Which machine learning model can be applied to the dataset?

In [ ]:
```
outliers = df[
    (df["Age_Numeric"] < lower_limit) |
    (df["Age_Numeric"] > upper_limit)
]

print("Number of outliers:", len(outliers))
```

```
outliers[["Age", "Age_Numeric"]]
```